Prince Mohammad bin Fahd University

College of Computer Engineering & Science

Department of Information Technology

Senior Design Project Proposal

AI-Based Building Evaluation and Contract Compliance System

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Advisor:

Sawsan Abdulrahman

Team Members:

Names:	**ID:**	**Major:**
Sadeem Alzahrani	202201067	Information Technology
Ghala Alshehri	202200942	Information Technology

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Introduction

The exponential growth of digital transformation in recent years has brought an increased reliance on data-driven systems for decision-making, communication, and innovation. Among the various forms of data, unstructured data such as free-text comments, online reviews, customer feedback, and detailed reports has become increasingly abundant. However, while structured data can be easily stored and analyzed using conventional database systems, unstructured data presents unique challenges due to its variability, length, complexity, and ambiguity. As a result, valuable insights often remain hidden within unorganized datasets, limiting the potential for organizations and researchers to fully exploit this information.

Artificial Intelligence (AI), particularly the fields of Natural Language Processing (NLP) and Machine Learning (ML), has emerged as a powerful solution to address this gap. By leveraging algorithms that can process human language and identify patterns, NLP-based systems have the potential to automatically classify, analyze, and summarize large volumes of textual data. This automation significantly reduces the need for manual intervention, minimizes errors, and accelerates the decision-making process. Moreover, the integration of NLP with advanced ML models has shown promising results in enhancing the accuracy of classification tasks and the interpretability of results, making it possible to bridge the gap between raw unstructured data and actionable insights.

The proposed project focuses on the design and development of an intelligent classification system that can automatically process unstructured comments and textual inputs. Unlike traditional keyword-based tools, this system will employ advanced NLP techniques combined with ML models to categorize comments more effectively, taking into consideration both semantic meaning and contextual relevance. This system can be particularly beneficial in domains such as real estate, where customer or resident feedback provides critical information about buildings, services, or facilities but is often overlooked due to its unstructured nature. By transforming raw feedback into structured categories, stakeholders will gain an efficient mechanism to evaluate performance, identify recurring issues, and improve services.

Another crucial aspect of this project is the emphasis on usability and adaptability. The solution is intended not only as a research prototype but also as a practical tool that can be deployed across multiple fields, including healthcare, education, and business. Its architecture will be designed to accommodate scalability, ensuring that the system remains effective even as the volume of data increases. Furthermore, the project will prioritize security and privacy, ensuring that sensitive data is protected through encryption and role-based access control. By combining technical innovation with practical usability, the proposed system contributes both academically and socially to the advancement of intelligent data analysis solutions.

Justification

The justification for this project arises from the significant gap that currently exists between the availability of unstructured data and the capacity of existing systems to process it meaningfully. In many industries, decision-making is based on incomplete or biased information because unstructured feedback is either ignored or analyzed manually, which is both time-consuming and error-prone. For instance, in real estate and property management, thousands of tenants and clients provide textual comments related to housing conditions, facilities, or customer service experiences. Without an automated tool to classify these comments, organizations cannot systematically identify patterns, prioritize issues, or develop strategies for improvement. This limitation not only affects operational efficiency but also impacts customer satisfaction and trust.

From a research perspective, the project is justified by its contribution to the ongoing exploration of NLP and ML methodologies in real-world applications. While NLP has made significant progress in language understanding, applying these models to specialized domains such as real estate or customer service still poses challenges due to domain-specific vocabulary, linguistic ambiguity, and varying comment structures. Developing a classification model tailored to these challenges will expand the academic knowledge base, providing insights into how NLP techniques can be adapted for niche applications. Furthermore, the evaluation of system performance using metrics such as accuracy, precision, recall, and F1-score will ensure the research maintains methodological rigor and aligns with established standards in AI evaluation.

In terms of societal and business impact, this project offers tangible benefits. By automating classification, the proposed system will reduce reliance on manual data processing, thus saving time and resources for organizations. It will also minimize human errors and inconsistencies, ensuring that results are more reliable and objective. For businesses, this translates into faster decision-making and improved responsiveness to customer needs. For end-users, such as tenants or clients, it ensures that their feedback is not only acknowledged but also acted upon systematically, leading to better services and increased satisfaction.

Another justification lies in the feasibility and sustainability of the project. By relying on open-source libraries and tools, such as Python's NLP and ML frameworks, the system can be developed within a manageable budget, making it accessible for academic research and practical deployment. At the same time, the project aligns with ethical and social considerations by prioritizing data security and user privacy. The system will be designed with compliance to international standards of data protection, ensuring trust and reliability.

In conclusion, the project is justified by its academic significance, practical utility, and social impact. It not only addresses a pressing gap in the processing of unstructured data but also proposes a solution that is feasible, sustainable, and adaptable across domains. The integration of NLP and ML into a robust classification system holds the potential to transform

raw textual data into meaningful knowledge, thereby bridging the divide between information abundance and actionable intelligence.

Literature Review

Study title and year	Focus	Key Findings	Limitations
Artificial Intelligence in Infrastructure Construction: A Critical Review (2024)	Conducted a systematic review of 300+ studies on AI use in construction, categorizing applications in design, monitoring, maintenance, and risk management.	Found that AI improves automation and safety; highlighted growth of NLP and multi-modal integration for compliance and decision-making.	lack real-world deployment, integration of text data, and enterprise scalability.
Artificial Intelligence in the Construction Industry: A Review of Present Status, Opportunities and Future Challenges (2021)	Reviewed 150+ studies on AI applications (ML, NLP, CV, robotics, expert systems) in construction to identify progress, opportunities, and gaps.	Found that AI improves productivity and safety; integration with BIM and IoT is key to smart, data-driven construction; NLP emerging for contracts.	Review-based; no implemented system; weak focus on data security, integration, and enterprise scalability.
Automated construction contract analysis for risk and responsibility assessment using natural language processing and machine learning (2025)	Used NLP and ML models (SVM, Random Forest, BERT) to automatically classify contract clauses into risk and responsibility categories.	Achieved >92% accuracy; automated extraction of risk and responsibility; reduced manual review time by 70–80%.	Limited dataset; focused only on risk/responsibility; no feedback or compliance linkage; single-user prototype.
AI-Enabled Decision Support Systems for Smarter Infrastructure Project Management in Public Works (2024)	Proposes a comprehensive AI-enabled Decision Support System (AI-DSS) framework for infrastructure project management, with emphasis on cost control, risk forecasting, compliance monitoring, and multi-stakeholder coordination in public works.	Meta-analysis of 178 studies shows AI-DSS improves forecasting accuracy, reduces cost overruns (15–25%), enhances schedule adherence, strengthens risk identification, and improves compliance and transparency when integrated with ERP/CRM/GIS and supported by explainable AI.	Primarily conceptual and framework-driven; lacks implementation of a concrete system. Focused mainly on U.S. public infrastructure, which may limit transferability. Relies on secondary studies rather than primary system deployment.

Table 1: Literature Review Summary

Detailed Requirements

1.1 System Overview

The proposed system, **AI-Based Building Evaluation and Contract Compliance System**, will be offered as a **Business-to-Business (B2B)** subscription platform for construction and real-estate companies. Each company subscribing to the platform will be assigned a dedicated **tenant workspace** that allows its authorized employees to manage projects, upload contracts, and analyze post-construction feedback using AI models.
Access is restricted to registered corporate users; self-registration is not allowed. User accounts are managed through a **company administrator** who provisions accounts via invitation.

The system supports **multi-tenant architecture**, ensuring data isolation and secure access per organization. Each company's data, users, and projects are logically separated from other tenants.

1.2 Functional Requirements

FR1: Multi-Tenant and Subscription Management

- The system shall allow the **Super Admin** to provision, suspend, or terminate tenant accounts for client companies.

- The system shall **define and enforce** subscription plans (Starter, Pro, Enterprise) that limit the number of projects, users, and monthly AI analyses.

- Tenants can upgrade or downgrade their subscription under predefined policies.

- Access to the system is by **invitation only**, ensuring that only verified employees can log in.

FR2: Authentication and Security

- The system shall support secure login with **email and password**, with optional **SSO (Single Sign-On)** integration using SAML or OIDC for Pro and Enterprise plans.

- **Two-Factor Authentication (2FA)** and **IP Allowlisting** shall be available for Enterprise customers.

- Passwords shall be hashed and transmitted via SSL/TLS.

- The system shall **enforce session timeouts** and **maintain audit logs** to ensure compliance with corporate security standards.

FR3: Role-Based Access Control (RBAC)

- The system shall **define** four main user roles.
    - **Super Admin**: platform-level management.
    - **Company Admin**: manages tenant users and projects.
    - **Project Manager**: oversees project execution and feedback analysis.
    - **Project Engineer**: handles assigned issues and technical reports.
- Each role shall have specific permissions and limited data visibility based on its scope.

FR4: Project and Contract Management

- Company Admins shall **create projects** and **assign Project Managers**.
- The system shall store all uploaded contracts with version control.
- Contracts may be provided in PDF or DOCX format, and the system shall **extract clauses** for AI analysis.

FR5: Feedback Recording and Management

- Project Engineers shall record pre-handover inspection feedback, including text, photos, or supporting documents.
- The AI engine shall auto-suggest a category (e.g., structural, safety, mechanical) during feedback submission.
- The system shall **link** each feedback entry to its respective project and contract.

FR6: Automated Feedback Classification and Prioritization

- The AI model shall classify feedback using **Natural Language Processing (NLP)** into categories such as *structural*, *mechanical*, *electrical*, *safety*, *usability*,

environmental, and *contractual*.

- The system shall detect **recurrent issues** and assign **priority levels** (Critical, Major, Minor) based on severity and frequency.

FR7: Contract Compliance Analysis

- The AI engine shall analyze uploaded contracts and compare their clauses against user feedback to determine accountability (contractor vs. owner).

- The system shall detect **non-compliance cases** and generate summarized compliance results.

- The system shall **apply text-matching and semantic similarity algorithms** for accurate mapping.

FR8: Contract Completeness Evaluation

- The AI system shall evaluate the completeness of each contract by checking for missing clauses or ambiguous statements.

- The system shall **generate a Contract Completeness Score** and **provide improvement suggestions**.

FR9: Issue Assignment and Resolution

- Project Managers shall assign detected issues to specific engineers.

- Engineers shall **acknowledge issues**, **add supporting evidence**, and **request reassignment** if needed.

- Once an issue is resolved, the Project Manager shall **review** and **approve** the resolution

FR10: Reporting and Digital Signatures

- The system shall generate detailed reports for compliance, feedback analysis, and project summaries.

- **Project Managers** shall approve reports technically, while **Company Admins** provide **official digital signatures** before publication.

- Reports shall be exportable in PDF format.

FR11: Dashboard and Visualization

- The system shall include interactive dashboards displaying project statistics, compliance scores, and issue trends.
- The system shall **update charts dynamically** based on user role and access rights.
- Tenants shall **view only their organization's data**.

FR12: Notifications and Alerts

- The system shall notify engineers and managers of pending actions, issue deadlines, and approval requests.

FR13: Data Retention and Export

- Upon subscription cancellation, the system shall retain tenant data for **90 days** with an additional **30-day backup grace period** before permanent deletion.
- Tenants can request **data export** (CSV/JSON + attachments) during this retention window.
- After this period, all tenant data shall be securely deleted and removed from backups.

1.3 Non-Functional Requirements

Constraint ID	Type	Description
C1	Technical	Each tenant's data **shall be logically isolated** with unique tenant IDs.
C2	Technical	The system shall process AI analysis requests within 30 seconds under normal load (≤100 concurrent users).

C3	Security	Passwords shall be stored as salted hashes, and all data transferred via HTTPS.
C4	Reliability	The platform shall maintain 99% uptime with daily automated backups.
C5	Performance	Dashboards shall load within 5 seconds for tenants with ≤1000 feedback records.
C6	Usability	The interface shall be responsive and accessible on desktop and tablet devices.
C7	Economic	Pricing plans shall limit features (users, projects, analyses) based on subscription tier.
C8	Environmental	Cloud hosting shall **use energy-efficient data centers** (ISO 14001 certified).
C9	Legal	The system must comply with data protection laws (e.g., GDPR-equivalent for enterprise clients).
C10	Safety	Access from unauthorized networks shall be blocked via IP allowlisting for sensitive tenants.
C11	Management	Tenants may upgrade/downgrade their plans with no data loss.
C12	Sustainability	The platform's business model shall support scalability and long-term maintenance.

Table 2: Project Constraints

1.4 Stakeholders and Use Cases

Actor	Type	Responsibilities / Major Use Cases
Super Admin	External	Provision and suspend tenants; manage subscription plans; enforce platform-wide security and auditing.
Company Admin	External	Manage company users and projects; approve and sign official reports; configure SSO and access controls.
Project Manager	External	Review feedback, upload contracts, run compliance analysis, assign issues, and approve technical reports.
Project Engineer	External	Resolve assigned issues, upload evidence, and track progress on the technical dashboard.

Table 3: Stakeholders and Major Use Cases

As shown in Table 3, each stakeholder has a clearly defined role within the system, ensuring proper separation of duties and secure access to system functionalities.

Figure 1 presents the system use case diagram, illustrating the main actors involved in the platform and their primary interactions with the AI-Based Building Evaluation and Contract Compliance System.

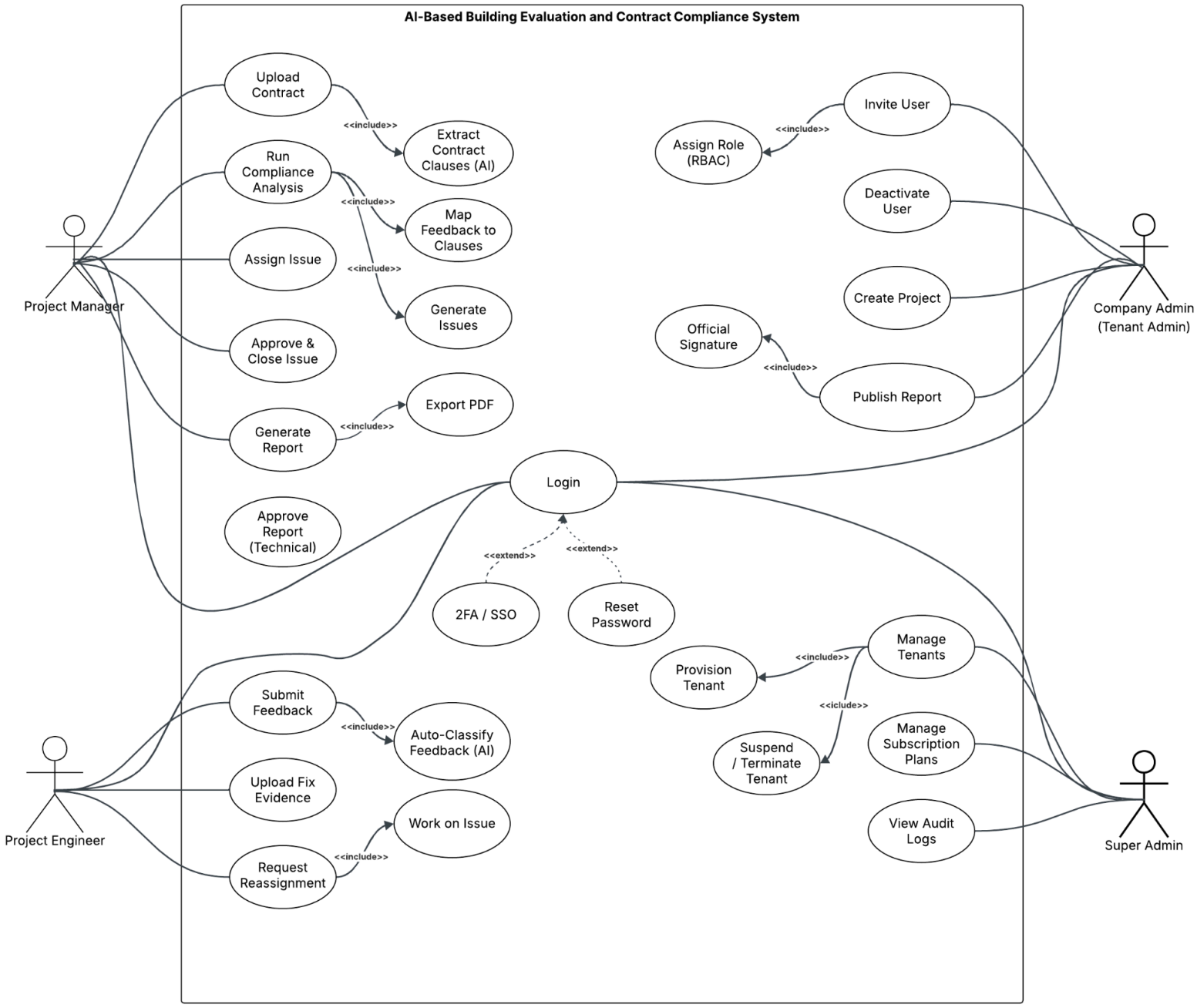

Figure 1: System Use Case Diagram

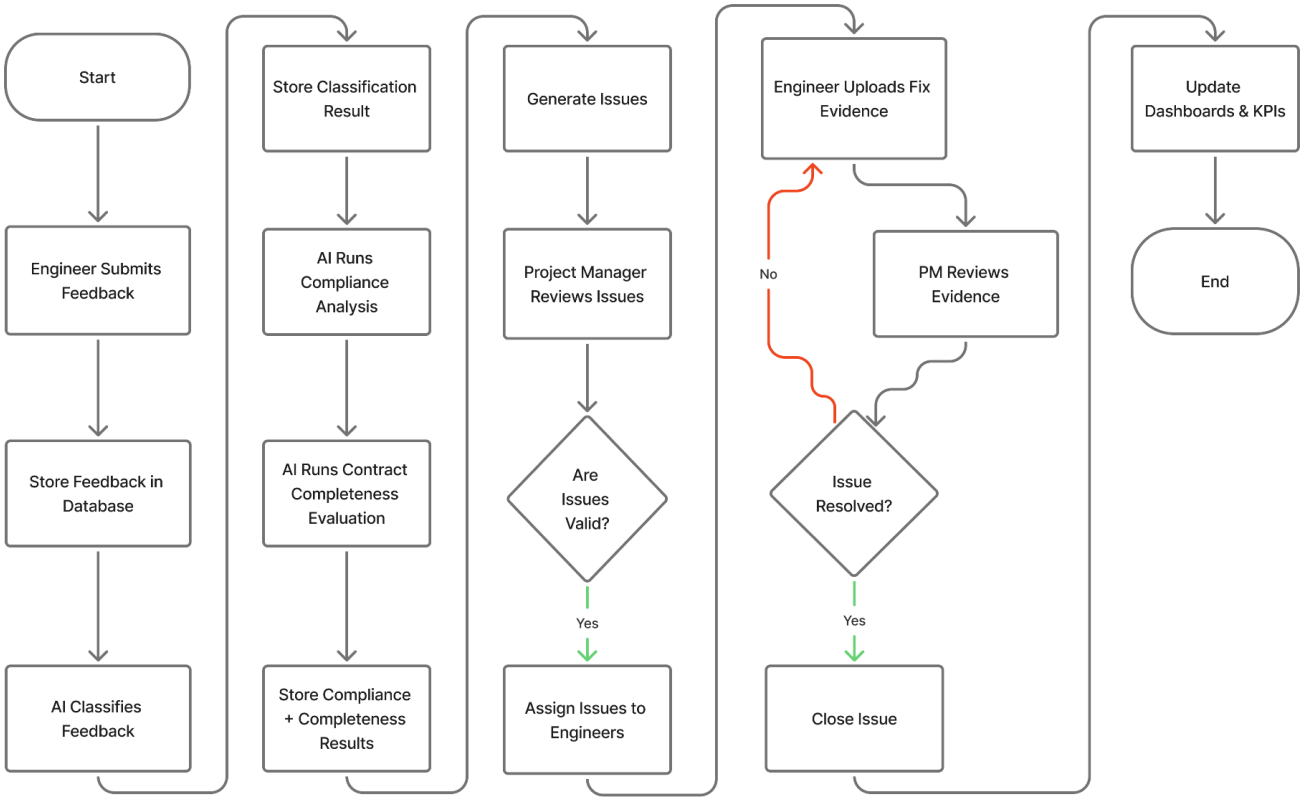

Figure 2: System Activity Diagram – Full Compliance Flow

Figure 2 illustrates the end-to-end workflow of the AI-Based Building Evaluation and Contract Compliance System. It demonstrates how feedback is recorded, classified, analyzed for compliance and completeness, transformed into issues, and finally validated and closed by the project team.

Figure 3 presents the sequence diagram describing the interactions between the Project Manager, User Interface, Backend, AI Engine, and Database during the compliance and completeness evaluation process.

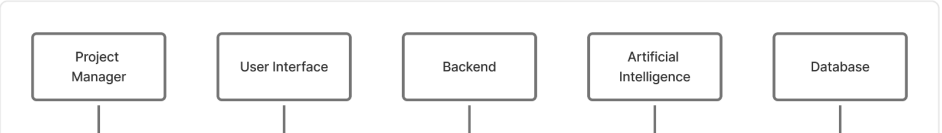

Figure 3: Full Compliance Analysis – Sequence Diagram

This diagram clarifies the real-time behavior of the system and ensures alignment with functional requirements FR5 to FR12.

System Mockup (High-Level User Interface Design)

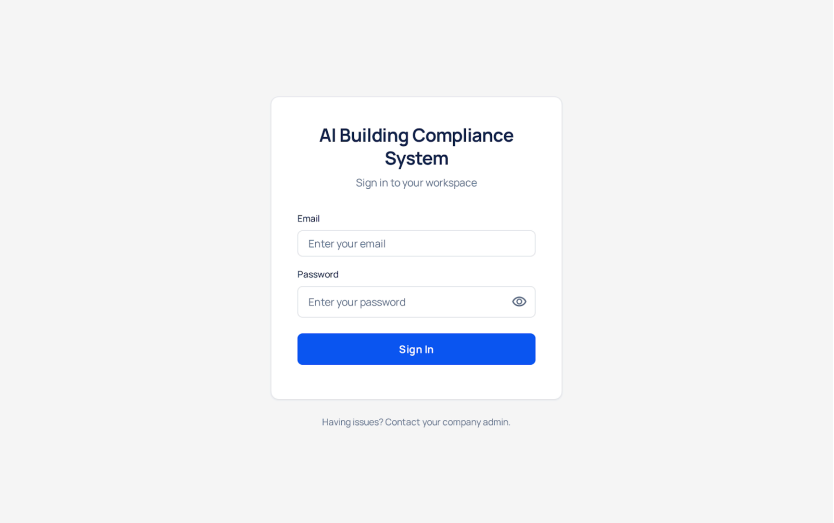

Figure 4: Login Page – Secure Workspace Access

The following screen represents the login interface used by all tenant users to securely access their workspace. The page includes email/password fields, password visibility control, and a prompt for contacting the company admin in case of login issues. This ensures secure authentication and a clean user experience.

Figure 5: Admin Dashboard – Project Compliance Overview

This dashboard provides a high-level overview of compliance performance, including total issues, critical issues, active projects, and overall compliance score. Visual trend charts help managers monitor issue progression over time.

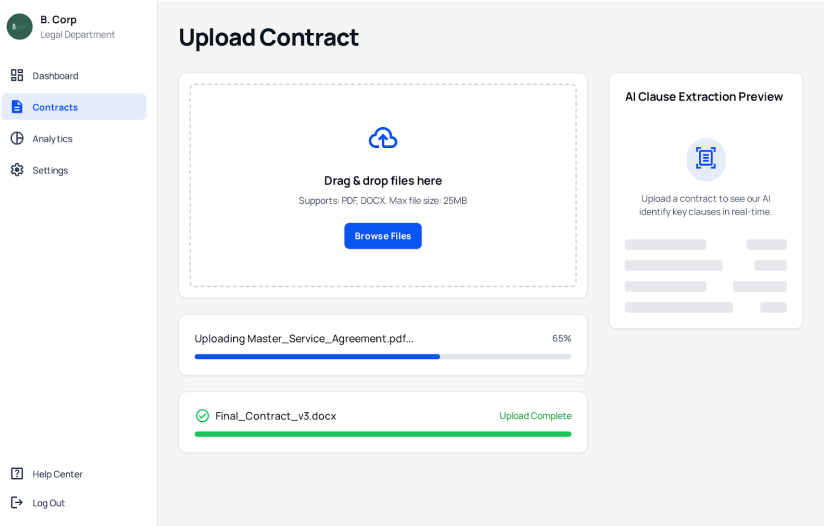

Figure 6: Contract Upload Interface

This interface allows legal or project managers to upload contract documents for AI processing. Supported formats include PDF and DOCX, with real-time progress tracking. A preview panel displays AI-extracted clauses once processing is complete.

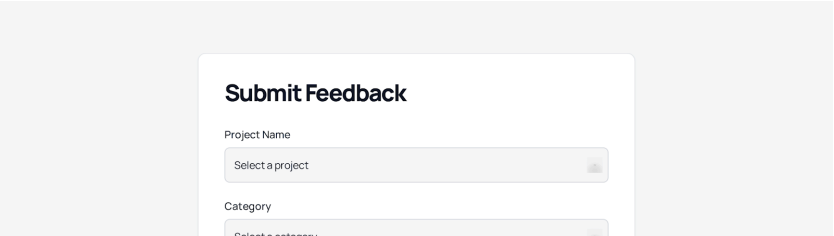

Figure 7: Feedback Submission Form

Engineers use this page to submit inspection feedback. The system auto-suggests the category using AI classification, and users can attach photos or documents to support their reports.

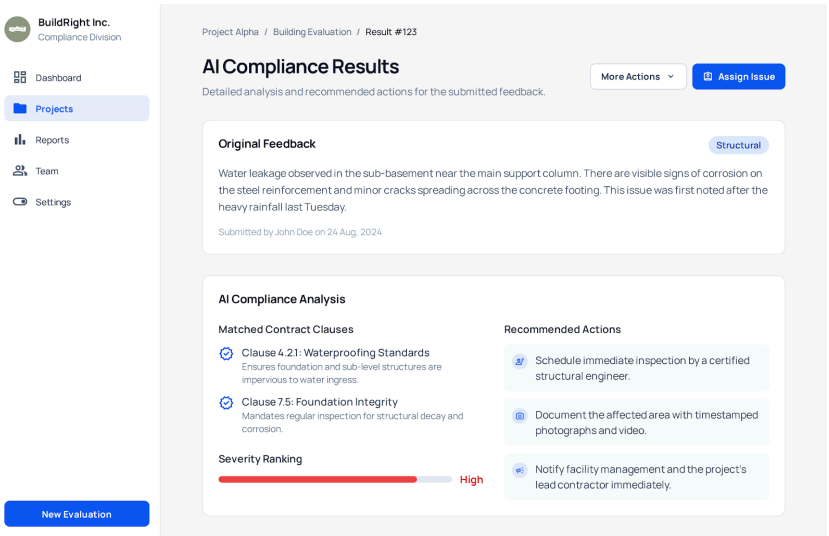

Figure 8: AI Compliance Results Page

This page displays the AI-generated compliance analysis, mapping user feedback to relevant contract clauses and providing recommended actions. Severity ranking helps prioritize critical issues.

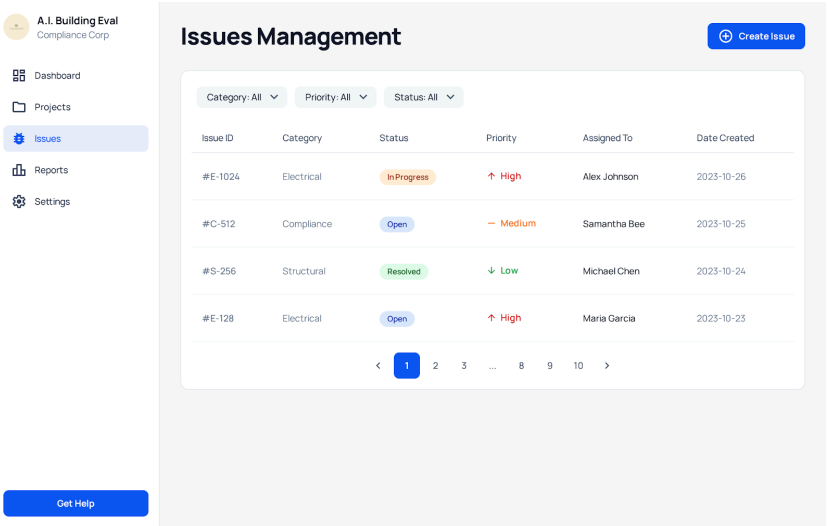

Figure 9: Issues Management Dashboard

This interface allows managers to review, filter, and track all issues across projects. Each row displays status, priority, category, assignment, and creation date to streamline task management.

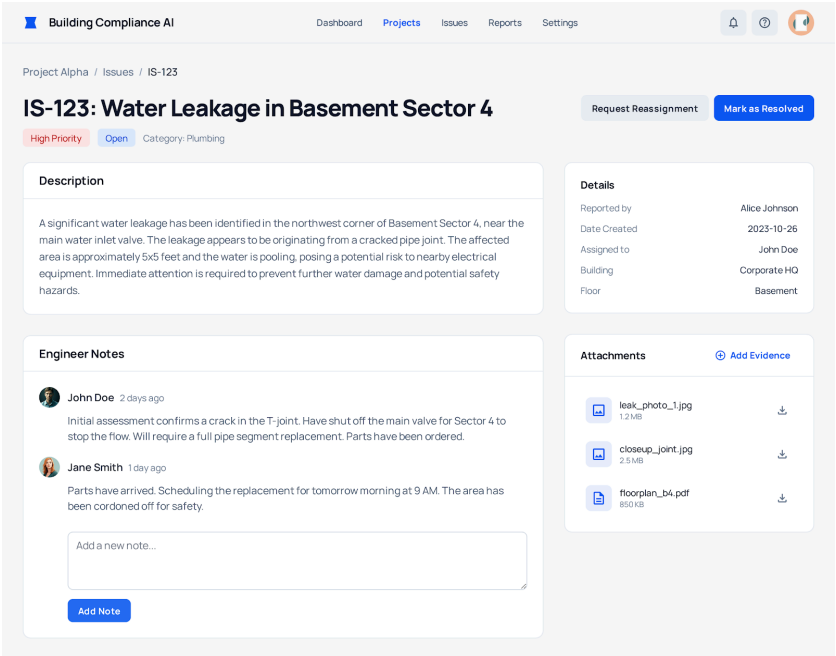

Figure 10: Issue Details and Engineer Notes

This page presents full issue details, including description, priority, assigned engineer, timestamps, and attachments. Engineers can log progress notes and upload additional evidence during issue resolution.

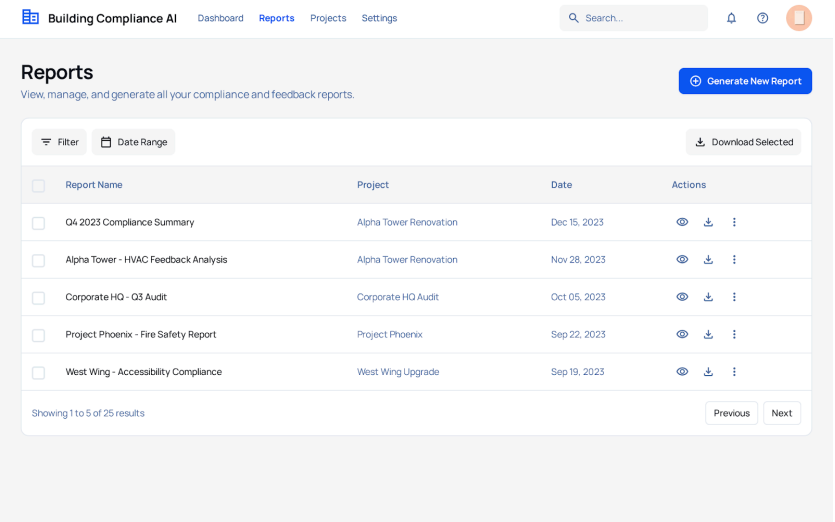

Figure 11: Reports Dashboard

The reports dashboard enables users to view, filter, and download compliance and feedback reports. Managers can generate new reports and access historical documentation for each project.

High-Level Architecture

The high-level architecture diagram provides an overview of the major system components and how they interact within the cloud environment. It illustrates the Client Layer, API Gateway, Application Server, AI Engine (classification, clause extraction, compliance, and completeness modules), and the Data Layer (database, file storage, backups). This architecture ensures scalability, security, and multi-tenant isolation across organizations.

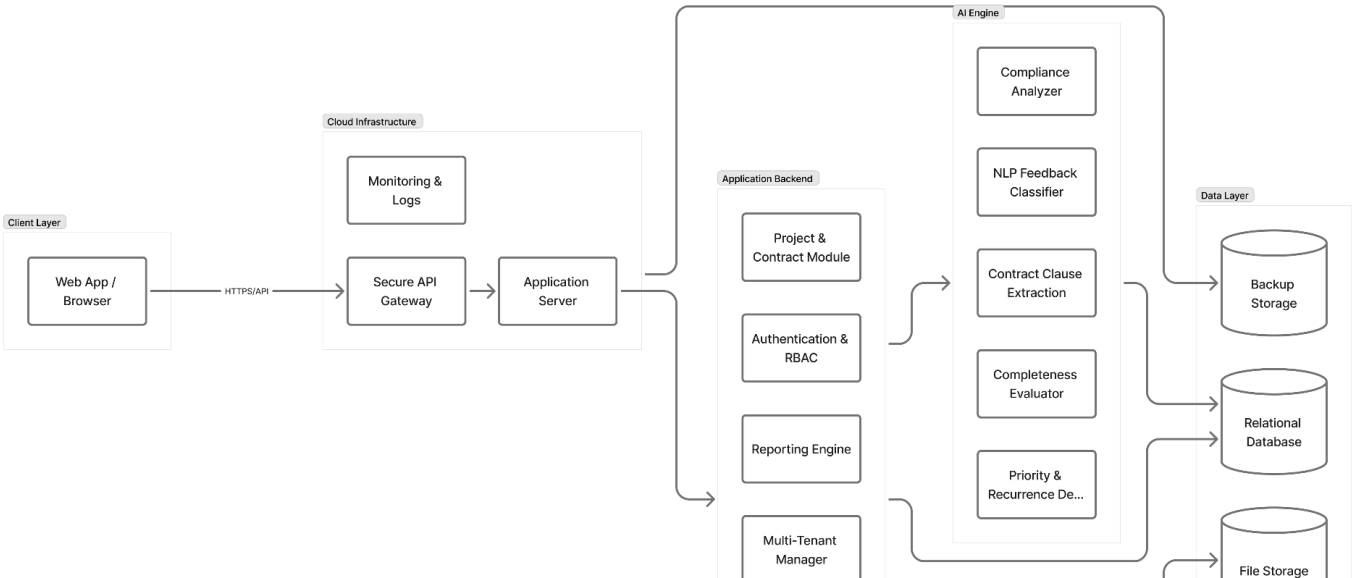

Figure 12: High-Level System Architecture

This architecture supports all functional and non-functional requirements, including multi-tenancy, secure access, role-based control, scalability, performance, and data protection. It provides the foundation upon which all system modules operate cohesively.

Security Requirements

The AI-Based Building Evaluation and Contract Compliance System must enforce strict security controls to protect sensitive contractual, technical, and corporate data across all tenant environments. All user authentication shall be secured through encrypted credentials using SSL/TLS, with passwords stored as salted cryptographic hashes. Role-Based Access Control (RBAC) will strictly regulate user permissions based on predefined roles to prevent unauthorized access to critical data and system functions. Enterprise tenants will be supported with advanced security features including Two-Factor Authentication (2FA), IP allowlisting, and Single Sign-On (SSO) integration. All uploaded contracts, inspection data, feedback, and reports shall be encrypted at rest and in transit. System activity logs and audit trails will be maintained for all user actions to support traceability, forensic analysis, and regulatory compliance. The platform must comply with international data protection regulations and apply the principle of least privilege to minimize exposure to insider threats, breaches, and data leakage.

Risk Assessment

- AI Model Accuracy Risk: The text classification models may not achieve the desired accuracy due to limited or biased training data.
  Impact: Low model accuracy can lead to misclassification of building evaluation data or contract clauses, resulting in incorrect compliance assessments or overlooked risks. This could erode stakeholder trust and lead to faulty decisions.
  Mitigation: Use diverse, high-quality training datasets and perform rigorous model validation. Continuously refine the AI models (e.g. retraining with new data) and implement human-in-the-loop review for critical outputs to ensure reliability.

- Data Privacy and Security Risk: Sensitive contract documents and evaluation data will be stored and processed in the cloud, raising concerns about unauthorized access or data breaches.
  Impact: A security breach could expose confidential contract information, leading to legal penalties and reputational damage. Loss of client trust and potential non-compliance with data protection laws are significant impacts.
  Mitigation: Enforce strong encryption for data at rest and in transit, and implement strict access controls and authentication. Regular security audits and compliance with standards like ISO/IEC 27001 ensure robust information security management. Also,

comply with privacy regulations (e.g. GDPR) to enforce data protection by design.

- Team Expertise and Training Risk: The project involves diverse areas (AI/ML, cloud DevOps, web development, security compliance). If team members lack experience in any of these areas, development could be slowed or implementation could be faulty.
  Impact: Skill gaps might result in suboptimal system design (e.g. insecure cloud setup or poorly tuned AI models) and require rework. The project could face delays as the team climbs the learning curve, and the final product quality may be lower without proper expertise.
  Mitigation: Mitigate by providing team training and workshops in critical technologies (such as Flask/Django, AWS security, or ML model development). If budget permits, bring in advisors or mentors for specialized areas (e.g. an AI specialist to guide model development). Adopt peer review practices so that more experienced members can help catch issues in unfamiliar domains.

- Regulatory and Compliance Change Risk: External regulations relevant to building compliance or AI usage may change during or after development. For example, new laws on AI ethics or data protection could impose additional requirements on the system.
  Impact: Changes in compliance standards could render the system non-conformant or require costly late-stage modifications. For instance, a new building code or contract law amendment might necessitate updates to the evaluation criteria, or an AI regulation might demand new transparency features.
  Mitigation: Stay informed on relevant regulatory developments throughout the project. Design the system with flexibility – e.g. using configuration files or modular rules engines for compliance criteria – so updates can be made without overhauling the core. Include legal/compliance experts in the loop to review the system against current laws. Additionally, build in features for audit trails and AI decision explainability to be prepared for stricter AI oversight requirements.

Risk(s)	**Potential Impact**	**Risk Mitigation**
AI model accuracy risk due to insufficient or biased training data	Misclassification of feedback or contract clauses, leading to incorrect compliance results and reduced system reliability	Use diverse and high-quality datasets, continuously retrain models, evaluate performance using standard metrics, and apply human review for critical outputs
Data privacy and security breach	Exposure of sensitive contracts, inspection data, or company information, causing legal and reputational damage	Encrypt data at rest and in transit, enforce Role-Based Access Control (RBAC), enable secure authentication, and follow ISO/IEC 27001 guidelines

System performance degradation under high user load	Slow response times, reduced usability, and negative user experience	Use scalable cloud infrastructure, optimize database queries, and apply load balancing and caching techniques
Incorrect mapping between feedback and contract clauses	Incorrect identification of responsibility and inaccurate compliance decisions	Combine semantic similarity techniques with rule-based validation and allow manual verification by project managers
Limited team expertise in AI or cloud deployment	Development delays, implementation errors, or reduced system quality	Conduct technical training, use online documentation and tutorials, and follow peer-review and supervisor guidance
Changes in regulations or compliance requirements	System non-compliance or need for major redesign	Design the system modularly, keep compliance rules configurable, and monitor regulatory updates throughout the project
Cloud service outage or infrastructure failure	Temporary system downtime and data access issues	Use reliable cloud providers, enable automated backups, and prepare disaster recovery and rollback plans
Poor quality or incomplete user feedback input	Reduced accuracy of AI analysis and reporting	Apply input validation, enforce structured feedback forms, and guide users using predefined categories

Table 4: Risk Assessment

Standards

The AI-Based Building Evaluation and Contract Compliance System will be developed in alignment with recognized international standards to ensure quality, security, and regulatory compliance.

- ISO/IEC 27001 (Information Security Management)
  This standard will guide the implementation of security controls to protect sensitive contract data, user information, and tenant-specific records through encryption, access control, and secure system operations.

- GDPR (General Data Protection Regulation)
  The system will follow GDPR principles to ensure lawful processing of personal

data, data minimization, user consent, and secure data storage, especially when handling contractual and inspection-related information.

- IEEE Software Engineering Standards
  IEEE standards will be followed to ensure proper software development practices, documentation, testing, and maintainability throughout the system lifecycle.

- NIST Cybersecurity Framework
  The NIST framework will be used as a reference for identifying, protecting, detecting, and responding to cybersecurity risks in the cloud-based, multi-tenant environment.

These standards ensure that the system is secure, reliable, and suitable for deployment in enterprise and regulatory-sensitive environments.

Tools and Processes

Tools

The system will be developed using Python as the main programming language due to its strong support for web development and artificial intelligence. A Python-based web framework such as Django or Flask will be used to implement backend functionality, authentication, and multi-tenant access control.

Natural Language Processing and machine learning will be implemented using libraries such as spaCy and scikit-learn, with optional use of pre-trained models from Hugging Face for advanced text analysis. A relational database (e.g., PostgreSQL) will be used for structured data, while cloud storage will handle uploaded contracts and files. The system will be deployed on a cloud platform to ensure scalability and availability.

Version control will be managed using Git, with automated testing and deployment supported through CI/CD tools.

Processes

The project will follow an Agile development process, using iterative cycles for design, implementation, testing, and refinement. AI models will be developed through repeated stages of data preparation, training, and evaluation. Continuous testing and validation will ensure system reliability and security. After deployment, the system will be monitored and improved based on user feedback and performance results.

Sub-Team Responsibilities and Team Structure

The project team is composed of two members with clearly defined roles and responsibilities. Tasks were distributed based on individual expertise to ensure efficiency, accountability, and successful project delivery.

Student 1: Sadeem Alzahrani (Team Leader)

Major: Information Technology

Expertise:

- System analysis and requirements engineering
- Software design and modeling (UML diagrams)
- AI-based system conceptualization
- UI/UX prototyping and system documentation

Responsibilities:

- Led the overall project planning and coordination activities.
- Defined the project justification, objectives, and system scope.
- Designed and documented the functional requirements and non-functional requirements.
- Developed all system diagrams, including Use Case Diagram, Activity Diagram, and Sequence Diagram.
- Designed the system mockups (high-level UI design) to represent the end-user experience.
- Created the high-level system architecture and ensured alignment between system components and requirements.

- Conducted continuous review of the project deliverables to ensure consistency, clarity, and compliance with senior design guidelines.

Student 2: Ghala Alshehri

Major: Information Technology

Expertise:

- Information security fundamentals
- Software standards and compliance
- Risk analysis and mitigation planning
- Development tools and process selection

Responsibilities:

- Defined and documented the security requirements, including authentication, authorization, encryption, and access control mechanisms.
- Identified and analyzed potential project risks and proposed appropriate mitigation strategies through a structured risk assessment.
- Selected and documented applicable international standards relevant to software quality, security, and data protection.
- Identified and justified the tools and development processes used in the project, including programming languages, frameworks, and AI libraries.
- Contributed to reviewing project documentation and provided feedback to improve accuracy, completeness, and alignment with academic requirements.

Week	Sections Covered	Work Completed	By
Week 1	Introduction + Justification	Defined project background, problem statement, objectives, scope, and justified the need for an AI-enabled DSS.	Sadeem, Ghala
Week 2	Detailed Requirements (Overall)	Defined system overview, main components, and functional requirements.	Sadeem
Week 3	Literature Review (Part 1)	Collected and reviewed core studies on AI, DSS, and project management	Ghala
Week 4	Literature Review (Part 2)	Analyzed literature findings, challenges, and identified research gaps	Ghala
Week 5	System Overview + Functional Requirements + Non-Functional Requirements	Defined system overview, main components, functional requirements, and Non- functional	Sadeem
Week 6	System Use Case Diagram	Designed use case diagram showing actors and system interactions.	Sadeem
Week 7	System Activity Diagram	Developed activity diagram representing system workflow.	Sadeem
Week 8	Sequence Diagram	Created sequence diagram illustrating system communication flow.	Sadeem
Week 9	System Mockup (UI Design)	Designed high-level user interface mockups for dashboards and reports.	Sadeem
Week 10	High-Level Architecture	Designed AI-DSS system architecture and integration flow.	Sadeem
Week 11	Security Requirements	Identified access control, data protection, and security mechanisms.	Ghala
Week 12	Risk Assessment + Tools	Analyzed risks, mitigation	Ghala

	and Processes	strategies, and identfies the tools	
Week 13	Final Review	Finalized documentation	Sadeem and Ghala

Table 5: Fall Project Plan and Timeline

Week	Milestone	Sadeem Responsibilities	Ghala Responsibilities
Week 1	Project planning & system requirements	Define functional requirements	Define non-functional & security requirements
Week 2	System architecture design	Design AI-DSS architecture	Design data flow & user roles
Week 3	Dataset collection & preparation	Collect contract documents	Prepare feedback and annotation data
Week 4	Data preprocessing	Clean and structure contract data	Prepare NLP-ready datasets
Week 5	NLP model development	Implement clause extraction	Implement text similarity & classification
Week 6	Machine learning model training	Train compliance classification model	Tune model performance & validation
Week 7	AI-DSS logic integration	Integrate ML/NLP outputs into DSS	Design decision rules & alerts
Week 8	Frontend / interface design	Dashboard layout & reports	User interaction & compliance views
Week 9	System integration & testing	Backend integration testing	Functional and usability testing
Week 10	Explainability & validation	Add explainable AI outputs	Validate results with sample cases
Week 11	Performance evaluation	Measure accuracy & efficiency	Analyze limitations & improvements
Week 12	Final system refinement	Fix bugs & optimize models	Improve UI and documentation
Week 13	Report Editing	Implementation & results sections	Discussion & limitations
Week 14	Final presentation & submission	Demo preparation	Slides & final review

Table 6: Spring Project Plan and Timeline

Expected Results

The expected outcome of this project is a functional and secure AI-Based Building Evaluation and Contract Compliance System capable of analyzing building inspection feedback and contractual documents using artificial intelligence techniques. The system is expected to successfully classify feedback, detect non-compliance cases, and generate structured compliance reports that support decision-making for construction and real estate organizations.

The project is expected to demonstrate effective integration between AI models and a multi-tenant web platform, providing accurate analysis, improved efficiency, and reduced manual effort in contract and building evaluation processes. Overall, the system is anticipated to be scalable, reliable, and suitable for real-world enterprise use.

Expected Difficulties & Challenges

One of the main challenges of the project is achieving high accuracy in AI-based text classification, as contract language and inspection feedback may vary significantly in structure and terminology. Limited or imbalanced training data may affect model performance.

Another challenge involves ensuring data security and privacy within a multi-tenant cloud environment, requiring careful implementation of access control and isolation mechanisms. Additionally, integrating AI components with the web application and managing system scalability may introduce technical complexity.

Time constraints and coordination between system development and AI model training may also present challenges; however, these issues can be mitigated through iterative development, testing, and continuous refinement.

References

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